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Integrating SIP and Dielectric Spectroscopy for Subsurface Property Assessment

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Abstract

This study aims to validate the effectiveness of spectral induced polarization (SIP) and dielectric spectroscopy for characterizing subsurface properties, specifically focusing on the very low and very high-frequency ranges, respectively. By comparing these two methods, the research aims to refine non-invasive techniques that enhance the reliability of subsurface properties across diverse settings. The study involves laboratory-based SIP measurements in the frequency range of 1mHz to 10kHz and dielectric spectroscopy measurements from 1MHz to 3GHz on core samples under controlled conditions. These measures are calibrated against known properties to evaluate and validate the two methods. Emphasis is placed on testing a range of rock types and saturation levels to explore the extent and limitations of these techniques. SIP and dielectric spectroscopy effectively capture variations in key subsurface properties. SIP measurements demonstrate strong potential in identifying surface area characteristics, while dielectric spectroscopy offers valuable insights into variations in porosity and permeability. However, findings indicate that certain measurements, specifically the imaginary part of the complex conductivity or permittivity, are sensitive to specific sample conditions, emphasizing the need for careful calibration to achieve broader applicability. This approach enhances the understanding of geophysical responses in rock formations, particularly in pure core plugs and carbonate rocks. These insights facilitate the broader application of these methods in assessing subsurface properties across diverse geological settings. This study contributes novel insights by demonstrating the integration of SIP and dielectric spectroscopy to a broader range of geological conditions. It offers a framework for integrating these methods into petrophysical assessments with greater generalizability, advancing non-invasive techniques for petroleum, hydrogeology, and environmental sectors.

Introduction

Accurate characterization of subsurface petrophysical properties is critical for numerous geoscientific and engineering applications, including reservoir evaluation, groundwater resource assessment, and environmental remediation. Properties such as porosity, saturation, permeability, and surface conductance govern the movement and distribution of fluids in porous media and are therefore central to any predictive

modeling or monitoring framework. While direct core analysis offers high-resolution measurements of these properties, such methods are invasive, limited in spatial coverage, and often constrained by sampling logistics, especially in inaccessible or heterogeneous subsurface environments. To overcome these challenges, geophysical methods have been developed as non-invasive alternatives to indirectly infer subsurface properties. Among these, Spectral Induced Polarization (SIP) (Kemna et al., 2012) and Dielectric Spectroscopy (DS) (Chen et al., 2022) offer promising capabilities due to their sensitivity to electrical and dielectric contrasts within porous media. These techniques operate over vastly different frequency ranges and are governed by distinct polarization mechanisms, making them complementary in the type and scale of information they provide. SIP typically spans the 1 mHz to 50 kHz range. It is susceptible to interfacial polarization effects at grain-fluid boundaries. In contrast, DS, which spans frequencies from 1 MHz to several gigahertz, is more responsive to molecular-scale dielectric relaxations and bulk matrix properties.

A significant gap in the current literature is the lack of integrated, frequency-spanning experimental designs that evaluate both SIP and DS responses across a range of lithologies and saturation states under controlled conditions. Additionally, despite the widespread use of Nuclear Magnetic Resonance (NMR) and laboratory liquid permeability measurements in core analysis, their integration with SIP and DS remains limited. NMR is a powerful technique that characterizes pore structure and fluid distribution (Ma et al., 2018) non-destructively, providing key parameters such as total porosity, T_2 relaxation distributions, and estimates of free versus bound water content. Similarly, liquid permeability measurements remain the industry standard for determining how easily fluids can flow through porous rocks. Incorporating these established methods alongside SIP and DS offer a unique opportunity to calibrate and validate the geophysical measurements against ground-truth petrophysical data. This study bridges this gap by conducting a comprehensive laboratory-based comparison of SIP and dielectric spectroscopy, calibrated against core plug NMR measurements and steady-state liquid permeability tests. The core samples span a range of lithologies, including clean sandstones, complex carbonates, and tight formations, under controlled saturation conditions. SIP measurements are conducted in the frequency range of 1 mHz to 10 kHz using a four-electrode setup, while dielectric spectroscopy covers the 1 MHz to 3 GHz range using precision impedance analyzers. NMR measurements are performed to quantify porosity and T_2 relaxation distributions, which are further used to estimate pore-size-related permeability trends. In parallel, standard steady-state permeability tests are conducted using brine under controlled flow and pressure conditions.

Each technique contributes a different perspective on the same system. SIP is particularly sensitive to the polarization effects associated with mineral surface area (Badhafere et al., 2023; Kirmizakis et al., 2024), ionic strength of pore fluids (Placencia-Gómez et al., 2016; Hördt et al., 2016), and surface conductance (Martin et al., 2020) phenomena. It provides insight into the electrochemical properties of the pore-fluid-solid interface, and the imaginary component of the complex conductivity is especially useful for identifying polarization mechanisms related to grain size, pore throat distribution, and surface roughness (Leroy et al., 2008). Dielectric spectroscopy, on the other hand, provides a more direct assessment of the dielectric permittivity and loss tangent, which are closely tied to volumetric water content, porosity, and the presence of bound versus free water. The high-frequency range of DS allows for the separation of different polarization processes, such as dipolar and interfacial relaxations, providing a nuanced view of pore fluid behavior under varying saturation conditions. When paired with NMR, the synergy becomes even more powerful. NMR T_2 relaxation distributions reflect the geometry and connectivity of the pore space. Short T_2 components are indicative of bound water associated with small pores or clays, while longer T_2 times correspond to free water in larger pore spaces. Since SIP and DS are also sensitive to fluid distribution and pore structure, NMR provides an ideal benchmark to interpret and constrain geophysical responses. Furthermore, permeability measured via steady-state flow is used to assess how well each technique correlates with fluid transport properties. The permeability-porosity relationship, often represented by empirical models such as the Kozeny-Carman equation, can be modified or enriched

using SIP-derived surface conductivity or DS-derived permittivity values, providing a more physics-based approach to permeability prediction.

One of the key research questions explored in this study is the extent to which SIP, DS, and NMR can jointly characterize pore-scale processes and how these inferences scale up to measurable macroscopic properties such as permeability. For instance, SIP may capture the development of polarization relaxation associated with partially saturated pores. At the same time, NMR quantifies the volumetric distribution of water, and DS captures dielectric relaxation shifts due to water content and mineral associations. A combined interpretation framework can then be established, where trends in SIP phase shift and imaginary conductivity are related to NMR-derived T_2 spectra. At the same time, DS permittivity trends are cross-validated against both NMR porosity and liquid permeability data. Initial findings demonstrate that SIP is highly effective in capturing surface-related phenomena, especially in partially saturated carbonate rocks where surface conductivity and membrane polarization are prominent. The frequency dependence of the phase shift and imaginary conductivity reflects changes in saturation and lithology, with strong correlations observed with surface area indicators derived from NMR. Meanwhile, dielectric spectroscopy proves highly sensitive to total porosity and water content, particularly at higher saturations. The dielectric constant and loss factors both increase with water saturation, in line with expectations from classical mixing models, but also reveal subtle effects of pore fluid distribution and mineralogy. Notably, the imaginary part of the complex permittivity in the gigahertz range exhibits high sensitivity to bound water content, supporting its use in differentiating between water types—a distinction also echoed in NMR measurements. Another important observation is that the permeability measured from steady-state flow shows a stronger correlation with combined DS and NMR parameters than with SIP alone. This likely reflects the fact that permeability is governed not only by surface effects (captured by SIP) but also by connected pore volume and geometry (which are better captured by DS and NMR). However, the inclusion of SIP-derived imaginary conductivity as a supplementary variable improves permeability prediction models, particularly in low-porosity or heterogeneous samples where traditional porosity-based predictions break down.

The broader implication of this work is the potential to establish a robust multi-method framework for petrophysical characterization that spans six orders of magnitude in frequency. This approach enables a comprehensive understanding of the electrical, dielectric, and hydrodynamic properties of geological materials. In practical terms, such a framework can be adapted for in situ applications, where SIP and high-frequency dielectric logging tools can be deployed alongside NMR logs and pressure-based permeability tests. The integration of laboratory-calibrated models enhances the interpretation of field-scale datasets and improves the reliability of subsurface models used in resource estimation, groundwater modeling, and environmental monitoring. Ultimately, this study contributes a novel methodology for integrating SIP, dielectric spectroscopy, NMR, and core permeability measurements into a unified experimental design. It highlights the complementary sensitivities of each technique and demonstrates how their joint application provides a more complete and calibrated understanding of subsurface petrophysical behavior. This integrated approach sets the stage for developing predictive models that are more accurate, generalizable, and physically grounded than those based on single-method datasets. By validating this approach across a range of rock types and saturation states, the study lays the groundwork for broader application in the energy, water, and environmental sectors.

Methods

Different core samples, composed of carbonate and limestone lithologies, were selected for integrated laboratory analysis. Before measurements, all samples were fully saturated in a 3% KCl solution to ensure comparable fluid conditions across techniques and minimize salinity-driven variability in polarization or dielectric behavior. The NMR measurements focused on acquiring T_2 relaxation distributions using a low-field spectrometer. Both incremental and cumulative T_2 curves were derived. The incremental distribution

revealed the relative contribution of different pore sizes, with short T_2 values corresponding to micropores and longer T_2 intervals associated with meso- and macropores. The cumulative curve indicated the total porosity of each sample, with particular emphasis on the dominant relaxation range that contributed most significantly to overall fluid-filled pore volume. A cutoff value of approximately 33 ms was employed to distinguish between bound and free fluids, a standard threshold widely adopted in carbonate petrophysical analysis. Permeability was evaluated through steady-state liquid flow measurements. Each sample was connected to a fluid injection apparatus under controlled pressure gradients, and permeability was calculated using Darcy's Law, accounting for the flow rate, fluid viscosity, and sample geometry. Results were averaged across multiple repetitions to reduce uncertainty.

Dielectric spectroscopy was applied in the frequency domain to quantify both the real (ϵ') and imaginary (ϵ'') components of complex permittivity. The real part ϵ' reflects the rock's capacity to store electrical energy. It is directly influenced by porosity and saturation, while the imaginary part ϵ'' indicates dielectric losses caused by fluid displacement and interfacial polarization. Measurements were conducted across a wide frequency range to capture both high-frequency capacitive behavior and low-frequency interfacial polarization. The rate at which ϵ' decayed with frequency was used as an additional diagnostic for pore connectivity and saturation state. SIP measurements were performed to obtain frequency-dependent complex conductivity in the low-frequency range (0.1 Hz to 10 kHz). The real component of conductivity (σ') relates to fluid pathways and matrix properties, while the imaginary part (σ''), expressed as phase shift, indicates the capacity for interfacial polarization and thus serves as a sensitive indicator of pore geometry and charge storage at interfaces. SIP measurements were interpreted in terms of both absolute magnitudes and spectral trends, with attention to how phase angle increased or decayed with frequency, signaling the degree and scale of pore-size-related polarization. Each method was applied to the same set of samples under identical saturation and environmental conditions.

Results

The NMR T_2 analysis (Fig. 1) revealed marked variability among the carbonate and limestone samples. Samples identified as carbonate (specifically samples 1 to 3) exhibited high cumulative porosity, typically exceeding 26%. Their T_2 distributions were dominated by macropores with relaxation peaks in the 500 to 1000 ms range, indicative of well-connected moldic and vuggy pores. These distributions suggested that a large proportion of the fluid volume resided in highly mobile domains, favorable for fluid storage and movement. In contrast, samples 4 and 5, which are predominantly composed of limestone, exhibited porosities of less than 24%. Their T_2 peaks occurred at shorter relaxation times, typically in the 10 to 100 ms range, signifying a dominance of mesopores and possible contributions from bound fluid. These results implied restricted connectivity and higher proportions of immobile water within the pore system.

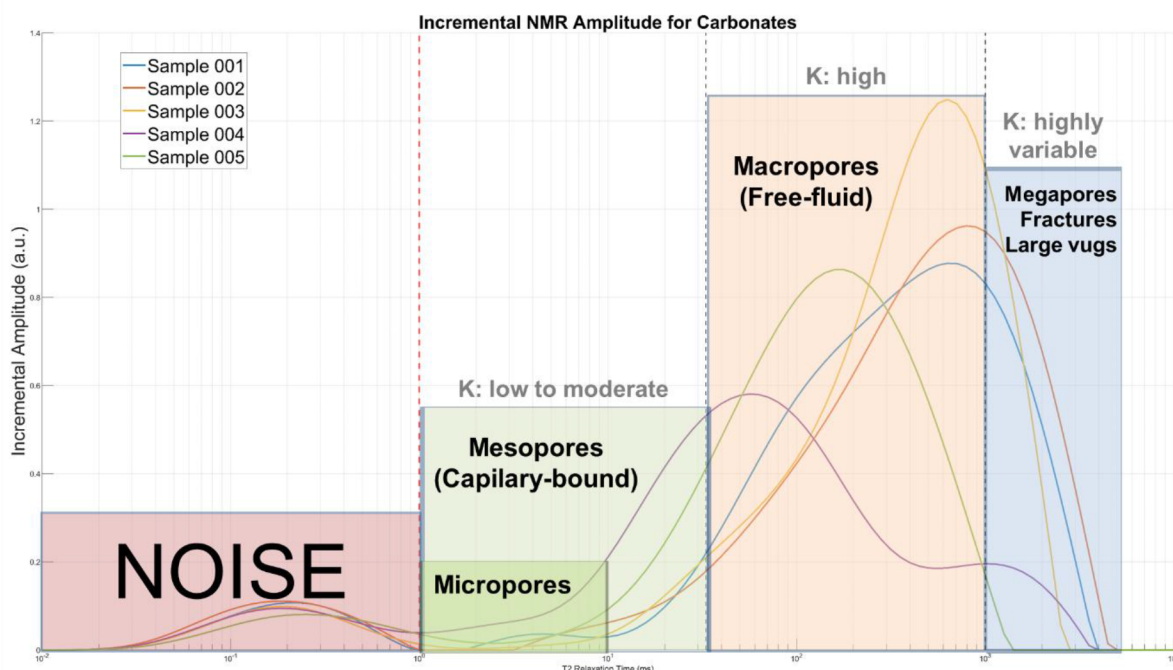


Figure 1—Incremental NMR T_2 amplitude distributions for five carbonate core samples, highlighting pore size classification and associated permeability regimes.

The measured liquid permeability values (Table 1) were consistent with the observed NMR trends. Samples with macroporous NMR profiles, such as sample 2 (115.38 mD) and sample 3 (111.82 mD), demonstrated high permeability, supporting the interpretation of effective fluid pathways and well-connected macropores. Conversely, samples 4 and 5 presented significantly reduced permeability values of 4.80 mD and 3.33 mD, respectively, despite having moderately developed pore systems. These discrepancies underscore the importance of pore-throat connectivity and tortuosity, rather than porosity alone, in determining fluid flow capacity. Sample 6, a limestone with intermediate porosity, exhibited a permeability of 7.28 mD, consistent with partially connected intercrystalline mesopores and suggesting heterogeneity in the pore network.

Table 1—Classification of real conductivity (σ) behavior across frequency ranges to pore system types and fluid connectivity (based on Markov et al., 2012).

Sample	Porosity (%)	Permeability (mD)	T_2 Peak (ms)	Interpretation
Carbonate 01	26.41	35.23	630.96	Dominant macroporosity, moderate connectivity, consistent with moderate permeability.
Carbonate 02	26.89	115.38	794.33	High macroporosity and connectivity explain high permeability.
Carbonate 03	27.90	111.82	630.96	Well-developed macroporosity, high permeability, connected vugs.
Carbonate 04	18.19	4.8	56.23	Mesoporosity with some micropores, low connectivity, low permeability.
Carbonate 05	22.22	3.33	177.83	Moderate microporosity, poor connectivity, and permeability are limited by throat constriction.

Dielectric spectroscopy results provided additional confirmation of the pore structure and fluid mobility trends. Samples with high permeability also exhibited high dielectric permittivity (ϵ') and conductivity

(σ), particularly in their lower sections, indicating saturated and well-connected macroporous zones. The vertical variation in dielectric response, with lower ϵ' and σ values observed at the top surfaces of some samples, suggested internal heterogeneity and partial saturation effects. In mesoporous and low-permeability samples, dielectric responses were more complex. The permittivity decayed sharply with increasing frequency, reflecting interfacial polarization effects consistent with tight pore networks and partially mobile water. In these cases, elevated ϵ' values at low frequencies did not translate into higher permeability, reaffirming the disconnect between pore volume and effective flow pathways in compacted or cemented limestone systems.

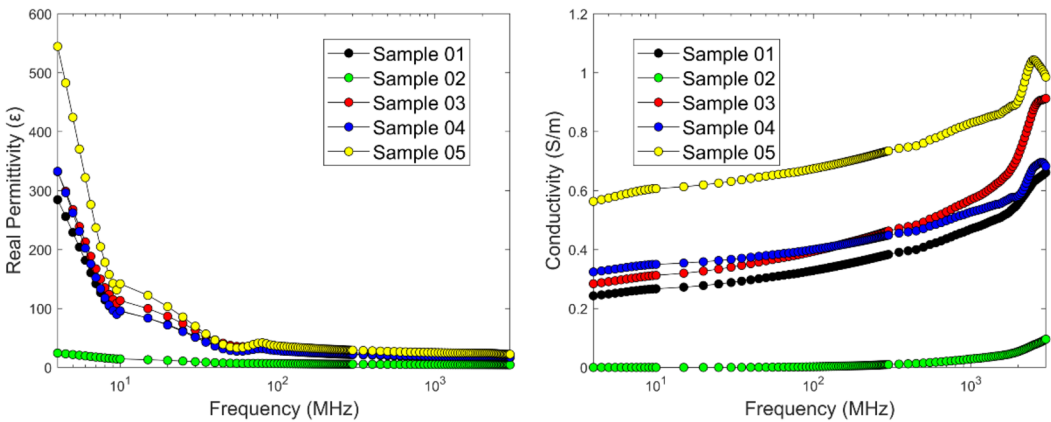


Figure 2—Frequency-dependent dielectric properties of five selected carbonate samples, showing real permittivity (ϵ') on the left and electrical conductivity (σ) on the right.

Table 2—Summary of petrophysical properties and interpretations for five carbonate core samples based on NMR-derived porosity, permeability measurements, and T_2 relaxation peak times.

σ Behavior (with frequency)	Pore System Type	Description
Very low σ (<0.01 S/m)	Tight / Dry matrix	Little or no free water, isolated pores, minimal ionic mobility.
Moderate σ (0.01–0.1 S/m)	Mesoporous / Intercrystalline	Some water content, moderate connectivity, and limited ionic pathways.
High σ (0.1–0.5 S/m)	Macroporous, partially connected	Presence of free water in larger pores, with moderate to good conductivity.
Very high σ (>0.5–1+ S/m)	Well-connected vuggy/saturated	High ionic mobility in continuous fluid paths, ideal for high-flow systems.

SIP measurements aligned with findings from both NMR and dielectric techniques. Samples with high NMR-derived macroporosity and high permeability demonstrated moderate SIP phase angles and real conductivity values that increased with frequency. These trends indicated well-developed polarization associated with interconnected macropores and high water saturation. The frequency-dependent behavior, characterized by a gradual phase increase and slight decay in σ' , confirmed the presence of broad pore-size distributions contributing to polarization. In contrast, SIP responses from tighter samples showed weak polarization signals, with low phase shifts and low σ' across the frequency spectrum. These responses were consistent with restricted fluid mobility and poorly connected pore structures. In one particular sample, despite the presence of visible macropores on NMR, the SIP response revealed minimal phase development and low conductivity, suggesting that pore throats were blocked or insufficiently conductive, an insight not easily discernible from NMR or dielectric data alone.

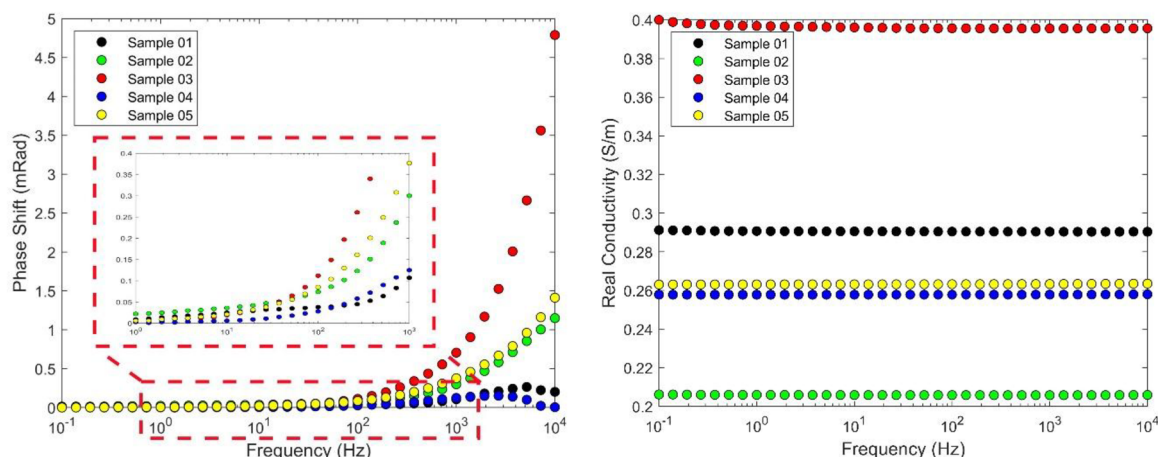


Figure 3—SIP results for five selected core samples measured at 0.1 Hz to 10 kHz. (Left) Phase shift (in mRads) as a function of frequency. An inset highlights the phase response between 1 and 1000 Hz. (Right) Real conductivity for the same samples across the measured frequency spectrum.

Overall, the consistency across methods affirmed the reliability of the multimethod approach. High-permeability samples showed agreement across all techniques—high NMR porosity, high ϵ' and σ , and moderate SIP phase and conductivity. Low-permeability samples were equally consistent, with tight pore networks evidenced by short T_2 relaxation times, weak dielectric responses, and suppressed SIP polarization. Furthermore, dielectric and SIP responses provided valuable distinctions between samples with similar porosity but differing connectivity, highlighting their importance as complementary tools to traditional core analysis. These observations emphasize the nuanced control of permeability by pore architecture, beyond bulk porosity, and demonstrate how integrated petrophysical characterization can resolve subtle heterogeneities within carbonate and limestone reservoirs.

Conclusions

This study provides a comprehensive assessment of subsurface petrophysical properties by integrating SIP, dielectric spectroscopy, NMR, and laboratory-measured liquid permeability. The multi-method approach allowed for high-resolution evaluation of pore geometry, saturation state, and permeability across a range of core samples. Each technique revealed different yet complementary aspects of the rock-fluid system, offering a more holistic view of the controlling mechanisms behind subsurface flow behavior. One of the most significant findings is the strong correlation between SIP results and NMR-derived T_2 relaxation distributions. Samples with long T_2 peaks, indicating macroporous, well-connected networks, also displayed enhanced SIP polarization responses, particularly within the 10 Hz to 1 kHz range. This demonstrates that SIP can effectively capture surface-related polarization processes associated with large and conductive pore throats. Conversely, samples with short T_2 peaks and low permeability exhibited weak SIP signals, confirming that limited ionic mobility and high tortuosity reduce polarization effects. This reinforces SIP's utility in identifying surface area and interface-related properties that are not readily apparent from porosity alone.

Dielectric spectroscopy, particularly the real component of permittivity (ϵ'), showed a strong linear correlation with NMR total porosity across all samples. High ϵ' values corresponded to elevated water content and porosity, making DS a reliable non-invasive proxy for volumetric water estimation. Furthermore, the rate of ϵ' decay with frequency provided insights into pore connectivity and saturation state: faster decay was associated with tighter pore systems, while stable or gradual decay indicated well-connected porosity. Notably, the imaginary component (ϵ'') of DS at higher frequencies (MHz to GHz range) exhibited sensitivity to bound water content, consistent with short T_2 components in the NMR spectra. This sensitivity enables discrimination between bound and free water, a distinction that is crucial

for understanding fluid retention, residual saturation, and hydrocarbon recovery potential. When comparing geophysical measurements with laboratory-determined permeability, a nuanced pattern emerged. DS conductivity, especially at mid frequencies, demonstrated a stronger correlation with measured permeability than SIP real conductivity (σ'). This is attributed to DS capturing both volumetric and interfacial conduction mechanisms that relate more directly to fluid transport capability.

A key insight from this study is the limited predictive power of porosity alone. Samples with similar NMR porosity values exhibited significantly different permeability due to variations in pore throat connectivity and architecture. For example, samples 02 and 03 had comparable porosities ($\sim 27\%$) but outperformed others in permeability due to their connected macroporous systems, a feature accurately reflected in both SIP and DS responses. In contrast, samples with moderate porosity but poor connectivity (e.g., sample 05) showed low permeability, weak SIP phase development, and sharp ε' decay, confirming the necessity of multi-parameter approaches for accurate reservoir characterization. Overall, the consistency among NMR, DS, and SIP measurements validates the robustness of the integrated methodology. Each technique captured distinct petrophysical phenomena: NMR illuminated pore-size distributions and fluid volumes; SIP provided insight into electrochemical and interfacial polarization processes; and DS captured dielectric behavior tied to saturation and mineralogy. Together, these tools resolve subtleties in fluid distribution and pore connectivity that are invisible to conventional methods. This work establishes a framework for scalable, multi-method petrophysical evaluation. It paves the way for more accurate permeability models, enhanced fluid typing, and improved reservoir heterogeneity, all of which are critical for hydrocarbon recovery, groundwater management, and environmental monitoring. The ability to span six orders of magnitude in frequency and tie electrical, dielectric, and hydrodynamic properties into one coherent interpretation makes this integrated approach highly valuable for future field and laboratory applications.

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